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ABSTRACT

International MBA programs face persistent challenges in delivering effective financial management education to linguistically and culturally diverse student cohorts. Traditional approaches often struggle to provide accessible, personalized support that promotes deep understanding and sustained engagement. This thesis investigates the integration of artificial intelligence (AI) tools based on large language models (LLMs) to enhance learning outcomes and examines the factors that influence student adoption in this context.

Using a mixed-methods design, the study combines a pre-test/post-test experiment with surveys grounded in an extended Technology Acceptance Model (TAM) and qualitative analyses of interaction logs and interviews. The quantitative component evaluates changes in performance and time efficiency, while the qualitative component explores interaction patterns with AI that are associated with improved learning. The TAM analyses assess reliability and validity of measurement constructs and estimate structural relationships predicting behavioral intention to use AI tools.

The results report improvements in learning performance and efficiency after AI-supported study, identify interaction strategies linked to better outcomes, and show that perceived usefulness and perceived ease of use are central predictors of adoption, with cultural and experiential factors moderating these relationships. Taken together, the findings inform pedagogical practice and tool design for AI-supported financial management education, offering actionable guidance for institutions serving diverse international student populations.

Keywords: Artificial Intelligence; Financial Management Education; Technology Acceptance Model; International MBA Students; Large Language Models

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CHAPTER 1 INTRODUCTION

1.1 Research Background

The financial sector is experiencing a transformative shift propelled by artificial intelligence (AI), which is redefining operational paradigms and strategic decision-making. AI technologies, including machine learning, natural language processing, and predictive analytics, are integral to applications such as risk assessment, fraud detection, algorithmic trading, and personalized customer service (**?**). For instance, AI-driven algorithms enable financial institutions to process vast datasets, uncovering patterns that enhance forecasting accuracy and optimize investment strategies (**?**). The market for AI in finance, valued at $9.45 billion in 2021, is projected to grow at a compound annual growth rate of 16.5% by 2030, underscoring its increasing significance (**?**).

This technological evolution necessitates a corresponding transformation in business education, particularly in financial management, to prepare future leaders for an AI-driven economy. International MBA students, who form a diverse cohort at institutions like Beijing International Studies University (BISU), face unique challenges that amplify the need for innovative educational approaches. BISU, renowned for its multidisciplinary programs in foreign languages, tourism, and business management, attracts students from varied cultural and educational backgrounds. These students often encounter language barriers and differences in prior academic exposure, which can impede their mastery of complex financial management concepts (**?**).

Traditional pedagogical methods, reliant on standardized curricula, may not adequately address the diverse needs of international students. Financial management courses, with their intricate terminology and quantitative rigor, pose particular difficulties for non-native speakers and those unfamiliar with Western financial systems. AI tools, especially large language models (LLMs) like ChatGPT, offer promising solutions by delivering personalized learning experiences. LLMs can provide real-time explanations, translate concepts into multiple languages, and adapt content to individual learning paces, thereby enhancing comprehension and engagement (**?**).

Empirical evidence supports the efficacy of AI in education. A study at Tsinghua University found that 98% of students improved their quiz scores by an average of 38.6 points after using LLMs for a history course, highlighting AI's potential to boost learning

outcomes (**?**). Similarly, AI applications in higher education, such as adaptive systems and intelligent tutoring, have been shown to personalize learning and improve student performance (**?**). In business education, institutions are increasingly integrating AI to equip students with skills relevant to modern industries, where AI automates tasks and provides data-driven insights (**?**).

The integration of AI in financial management education at BISU could address the specific needs of international students while aligning with global trends in business education. However, the effectiveness of AI tools in this context and the factors influencing their acceptance among students remain underexplored. This study seeks to fill this gap by investigating AI's impact on learning outcomes and adoption dynamics, contributing to both educational practice and academic research.

1.2 Research Purpose and Significance

1.2.1 Research Purpose

This study has two primary objectives. First, it aims to assess the effectiveness of AI tools, specifically LLMs, in enhancing the learning outcomes of international MBA students in financial management at BISU. Through an experimental design, students will complete a financial management quiz, interact with AI tools, and retake the quiz to measure performance improvements. Second, the study seeks to identify factors influencing students' acceptance of these tools, using the Technology Acceptance Model (TAM) to examine constructs such as perceived ease of use and usefulness. This dual focus provides a holistic understanding of AI's educational impact and adoption barriers.

1.2.2 Research Significance

The research offers substantial contributions to academia and practice. Academically, it advances the literature on AI in education by providing empirical evidence on its application in financial management for a diverse student population. By employing TAM, the study enriches insights into technology acceptance in higher education, particularly in multicultural settings (**?**). The findings could inspire further research into AI's role in addressing educational disparities.

Practically, the study provides actionable insights for BISU and similar institutions. Demonstrating AI's potential to enhance learning outcomes can guide the development of tailored teaching strategies that accommodate international students' needs. Understand-

ing acceptance factors will inform the design of user-friendly AI tools, ensuring effective integration into curricula. As AI becomes ubiquitous in finance, equipping students with relevant skills enhances their employability and readiness for industry demands (**?**). This research thus bridges academic training and professional practice, fostering innovative educational approaches.

1.3 Framework

To provide a comprehensive overview of the research process spanning all six chapters, Figure **??** presents the framework of this study. This diagram illustrates the research approach (Chapter **??**), content (Chapters **??** and **??**), and methods (Chapter **??**), with results and analysis and conclusions addressed in Chapters **??** and **??**, respectively.

1.4 Structure of the Thesis

This thesis is organized into six chapters, each contributing to the investigation of AI adoption in education and financial management.

- **Chapter 1** introduces the research background, outlines the problem statement, objectives, research questions, and highlights the significance of exploring AI integration in education, with a focus on financial management instruction.
- **Chapter 2** reviews existing literature on the use of artificial intelligence in education and financial management, identifies theoretical and empirical gaps, and sets the foundation for the proposed research framework.
- **Chapter 3** presents the theoretical framework underpinning the study, with an emphasis on the Technology Acceptance Model (TAM) and its relevance to educational technology adoption.
- **Chapter 4** describes the research methodology, including the research design, data collection methods, instruments, and analysis strategies used to investigate user perceptions and behavioral intentions.
- **Chapter 5** presents the empirical *results and analysis*, including quantitative outcomes from the experiment, qualitative insights from interviews/focus groups, and a comprehensive TAM-based analysis (measurement validation and structural model testing).
- **Chapter 6** provides *discussion and conclusion*, interpreting the findings, outlining

theoretical and practical implications, acknowledging limitations, and proposing directions for future research.

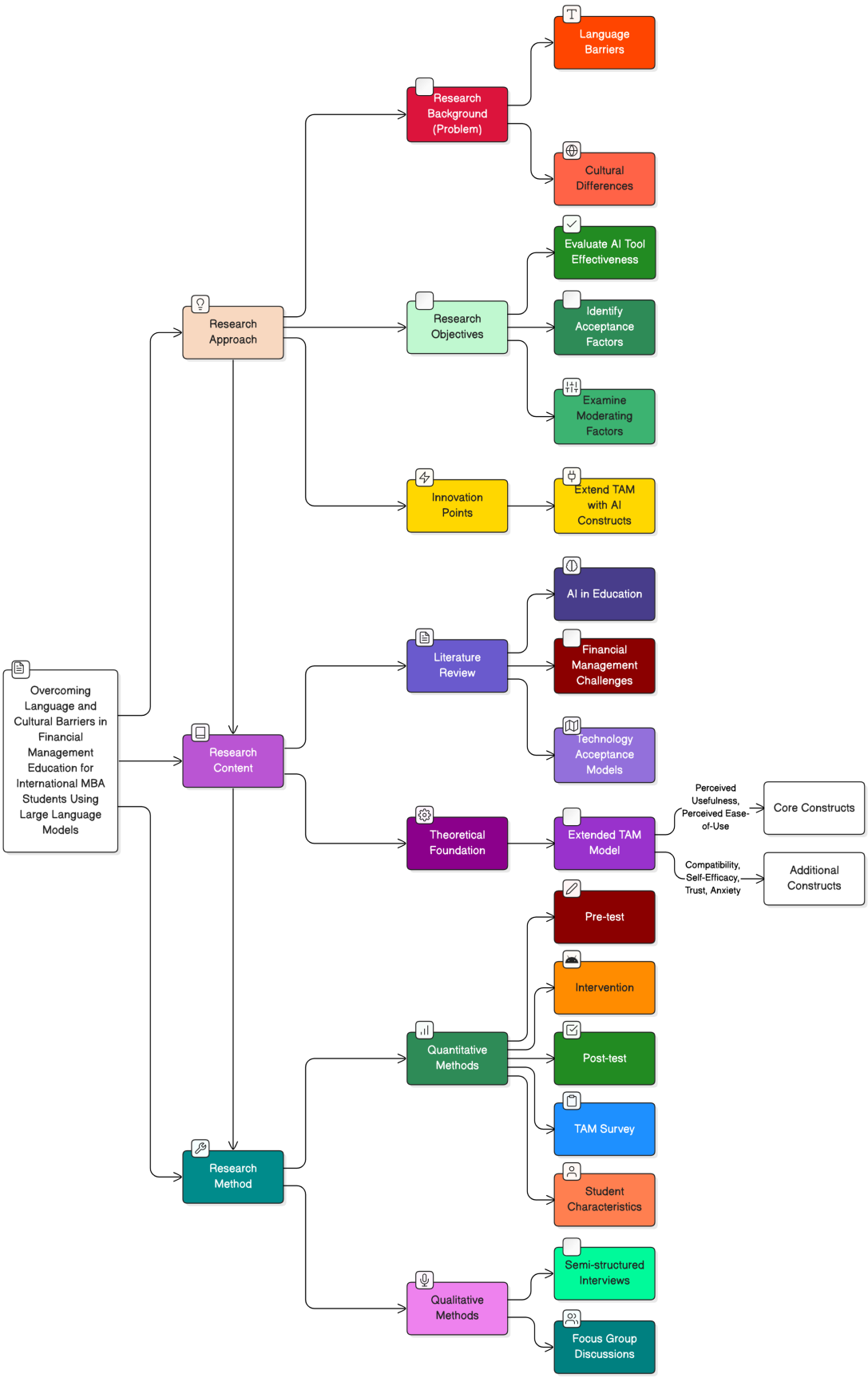

Figure 1.1 Framework of the Research Process

CHAPTER 2 LITERATURE REVIEW

2.1 Definitions of Key Concepts

Artificial Intelligence (AI) refers to the development of computer systems that can perform tasks requiring human intelligence, such as learning, decision-making, and perception (**?**). In educational contexts, AI encompasses tools that automate administrative tasks, personalize learning experiences, and provide data-driven insights to enhance pedagogical outcomes (**?**).

Large Language Models (LLMs) are advanced AI systems trained on vast datasets to process and generate natural language, enabling applications like automated tutoring and content creation (**?**). These models, such as GPT-4, have shown remarkable capabilities in understanding context and producing coherent responses, making them valuable in educational settings.

Financial Management Education involves the systematic instruction of concepts and skills related to managing financial resources, including budgeting, investment strategies, and risk assessment (**?**). Within MBA programs, this discipline equips students with the analytical tools necessary for strategic decision-making in corporate and financial environments.

The Technology Acceptance Model (TAM) is a widely used framework that predicts user adoption of technology based on two primary constructs: perceived usefulness, the degree to which a person believes a technology enhances performance, and perceived ease of use, the extent to which it is considered effortless to use (**?**). TAM is particularly relevant for assessing how students and educators embrace AI tools in academic settings.

2.2 Introduction to Technology Acceptance Theories

Technology acceptance theories provide frameworks for understanding how and why individuals adopt new technologies. These theories have evolved over decades, beginning with the Theory of Reasoned Action (TRA) in the 1970s, progressing to the Theory of Planned Behavior (TPB) in the 1980s, and culminating in the Technology Acceptance Model (TAM) and its extensions. Each theory builds upon its predecessors, addressing limitations and incorporating additional factors to enhance explanatory power (**?**).

The theoretical framework for this study is primarily based on the Technology Acceptance Model (TAM), which has been widely validated across various technological contexts, including educational settings (**?**). However, to comprehensively address the unique aspects of AI adoption in international educational contexts, this framework incorporates elements from related theories and extends TAM with constructs specific to AI and cross-cultural learning environments.

2.2.1 Definition of Technology Acceptance

Technology acceptance refers to the demonstrable willingness within a user group to employ technology for the tasks it is designed to support (**?**). In educational contexts, acceptance encompasses both behavioral intention to use technology and actual usage behavior (**?**). For AI tools in financial management education, acceptance involves students' willingness to engage with these tools for learning complex financial concepts, solving problems, and enhancing their educational experience.

Acceptance is distinguished from mere adoption, as it implies not only initial use but also continued engagement and integration into learning practices (**?**). This distinction is particularly relevant for AI tools, which may require sustained interaction to deliver personalized benefits. Furthermore, acceptance in educational settings is influenced by both individual factors (e.g., prior experience, personal innovativeness) and contextual factors (e.g., institutional support, peer influence) (**?**).

2.2.2 Importance of Technology Acceptance in Educational Settings

The importance of technology acceptance in educational settings cannot be overstated, particularly for innovations like AI that promise transformative benefits. Research indicates that technological innovations in education often fail not due to technical inadequacies but because of insufficient user acceptance (**?**). This phenomenon is especially pronounced in higher education, where faculty and student resistance can significantly impede technological integration (**?**).

For international MBA students studying financial management, acceptance of AI tools is particularly crucial for several reasons. First, these students face unique challenges, including language barriers and diverse educational backgrounds, which AI can help address through personalization (**?**). Second, financial management courses involve complex concepts and quantitative analyses that benefit from the interactive support AI can provide (**?**). Third, as future business leaders, these students will operate in an increas-

ingly AI-driven financial sector, making familiarity with such technologies professionally valuable (**?**).

Empirical studies have demonstrated that technology acceptance significantly predicts learning outcomes in technology-enhanced environments. For instance, **?** found that higher levels of acceptance were associated with improved performance and satisfaction among students using e-learning systems. Similarly, **?** reported that acceptance factors explained 67% of the variance in learning outcomes for educational technologies. These findings underscore the importance of understanding and fostering acceptance of AI tools in financial management education.

2.3 Theoretical Evolution of Technology Acceptance

The theoretical understanding of technology acceptance has evolved significantly over the past five decades, reflecting changing technological landscapes and deepening insights into human-technology interaction. This evolution provides a rich foundation for examining AI acceptance in educational contexts.

2.3.1 Theory of Reasoned Action (TRA)

The Theory of Reasoned Action, developed by **?**, represents one of the earliest systematic attempts to explain human behavior through attitudinal and normative factors. TRA posits that behavioral intention, the immediate antecedent of actual behavior, is determined by two primary factors: attitude toward the behavior and subjective norms (**?**).

Attitude toward behavior refers to an individual's positive or negative evaluation of performing a particular action, formed through an assessment of the behavior's expected outcomes and the value placed on those outcomes (**?**). Subjective norms reflect perceived social pressure to perform or not perform the behavior, influenced by normative beliefs about what important referents think and the motivation to comply with these referents (**?**).

In the context of AI acceptance in education, TRA suggests that students' intention to use AI tools would be influenced by their attitudes toward these tools (e.g., whether they believe AI will enhance their learning) and subjective norms (e.g., whether peers and instructors encourage AI use). However, TRA has been criticized for assuming that behavior is under volitional control and for neglecting factors that might constrain or facilitate behavior regardless of intention (**?**).

Despite these limitations, TRA provides valuable insights into the role of attitudes and social influence in technology acceptance. Studies applying TRA to educational technology have found that positive attitudes and supportive social environments significantly predict adoption (**?**).

As illustrated below in Figure **??**, the TRA model emphasizes the influence of attitudes and subjective norms on behavioral intention.

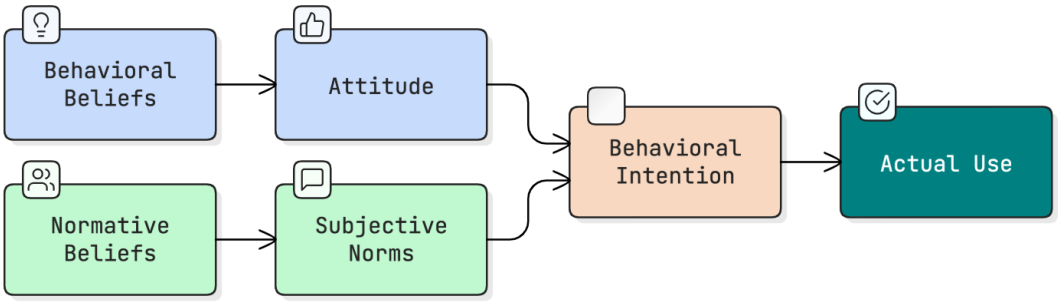

Figure 2.1 Theory of Reasoned Action (TRA) Model (Fishbein & Ajzen, 1975)

However, TRA's assumption that all behavior is under complete volitional control led to criticisms that it could not fully explain behavior in real-world contexts, particularly where external constraints are present. To address these limitations, the Theory of Planned Behavior (TPB) was introduced.

2.3.2 Theory of Planned Behavior (TPB)

Addressing limitations of TRA, **?** developed the Theory of Planned Behavior by adding perceived behavioral control as a third determinant of behavioral intention. Perceived behavioral control refers to an individual's perception of the ease or difficulty of performing a behavior, reflecting past experiences and anticipated obstacles (**?**).

TPB posits that individuals are more likely to form intentions and perform behaviors when they evaluate them positively, perceive social pressure to perform them, and believe they have the capability and resources to do so (**?**). This addition makes TPB particularly relevant for technological contexts, where users' perceptions of their ability to use technology effectively can significantly influence adoption decisions.

In educational settings, TPB has been applied to explain technology adoption among students and faculty. For instance, **?** found that perceived behavioral control was a significant predictor of teachers' intentions to use technology, while **?** demonstrated that all three TPB constructs influenced students' acceptance of mobile learning.

TPB also acknowledges that behavioral, normative, and control beliefs may vary across populations and contexts (**?**). This recognition is particularly relevant for international students, whose beliefs may be influenced by cultural factors and prior educational experiences. For example, students from collectivist cultures might place greater emphasis on normative beliefs, while those with limited exposure to advanced technologies might have lower perceived behavioral control (**?**).

As shown below in Figure **??**, TPB extends TRA by incorporating perceived behavioral control, thereby improving its predictive capability for non-volitional behaviors.

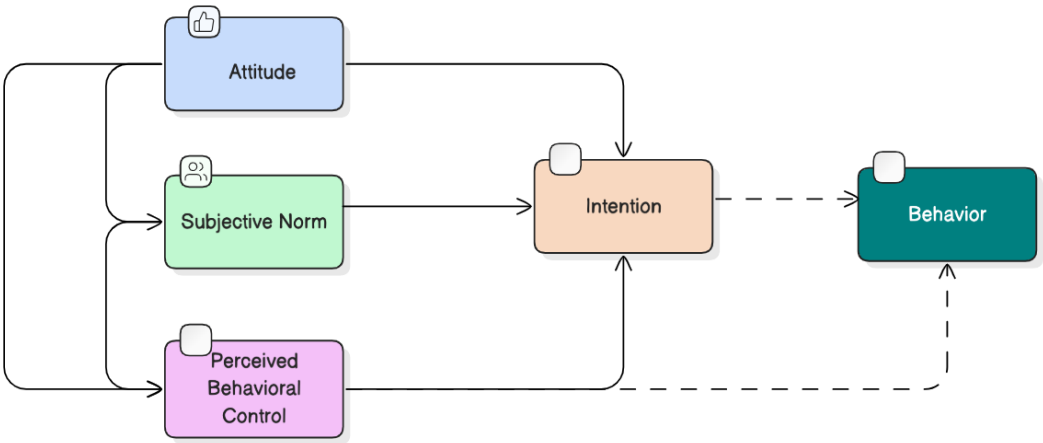

Figure 2.2 Theory of Planned Behavior (TPB) Model (Ajzen, 1991)

While TPB provided a more complete understanding of intentional behavior, researchers sought a more technology-specific model to predict user acceptance more precisely. This led to the development of the Technology Acceptance Model (TAM).

2.3.3 Technology Acceptance Model (TAM)

The Technology Acceptance Model, introduced by **?**, represents a significant adaptation of TRA specifically for information systems contexts. TAM posits that two primary beliefs—perceived usefulness and perceived ease of use—determine an individual's attitude toward using technology, which in turn influences behavioral intention and actual usage (**?**).

Perceived usefulness is defined as "the degree to which a person believes that using a particular system would enhance his or her job performance," while perceived ease of use refers to "the degree to which a person believes that using a particular system would be free of effort" (**?**). TAM suggests that perceived ease of use influences perceived usefulness,

as technologies that are easier to use are more likely to be perceived as useful.

TAM has been extensively validated across various technologies and user populations, consistently explaining a substantial portion of variance in usage intentions and behavior (**?**). In educational contexts, TAM has been applied to explain acceptance of e-learning systems (**?**), mobile learning (**?**), and educational software (**?**), among others.

For AI tools in financial management education, TAM suggests that international MBA students would be more likely to accept these tools if they perceive them as enhancing their learning (usefulness) and requiring minimal effort to use (ease of use). However, while TAM provides a parsimonious explanation of technology acceptance, it has been criticized for excluding social and contextual factors that may be particularly relevant in educational settings (**?**).

As depicted in Figure **??**, TAM offers a streamlined yet powerful framework for predicting technology adoption by focusing on users' beliefs about usefulness and ease of use.

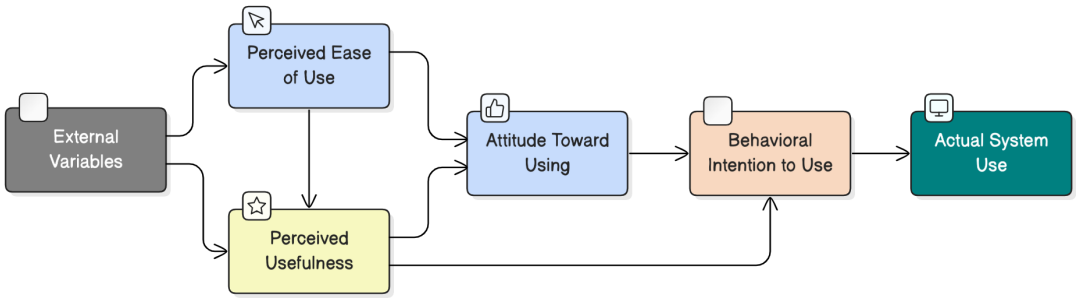

Figure 2.3 Technology Acceptance Model (TAM) (Davis, 1989)

Recognizing these limitations, researchers have proposed various extensions to TAM. TAM2 (**?**) incorporates social influence processes and cognitive instrumental processes as determinants of perceived usefulness. The Unified Theory of Acceptance and Use of Technology (UTAUT) (**?**) integrates elements from eight models, including TAM, TPB, and TRA, to provide a more comprehensive explanation of technology acceptance.

2.4 AI Applications Across Various Industries

AI has transformed numerous sectors beyond finance, demonstrating versatile applications and significant impacts. In healthcare, AI systems analyze medical images with accuracy comparable to specialists, with studies showing diagnostic precision rates of 84-95% for certain conditions (**?**). AI-powered predictive analytics also help identify patients

at risk of readmission, reducing hospital return rates by 10-18% in some implementations (**?**).

In manufacturing, AI-driven predictive maintenance has revolutionized operations by forecasting equipment failures before they occur. Companies implementing these systems report maintenance cost reductions of 10-40% and downtime decreases of 30-50% (**?**). Additionally, AI-powered quality control systems can detect defects with greater precision than human inspectors, improving product quality while reducing labor costs.

The retail sector has embraced AI for personalized customer experiences and inventory management. Recommendation engines powered by machine learning algorithms have increased sales by 5-15% for online retailers (**?**). Meanwhile, AI-based demand forecasting has helped reduce inventory costs by up to 20-30% while minimizing stockouts (**?**).

In transportation and logistics, AI optimization algorithms have improved route efficiency by 5-10%, resulting in significant fuel savings and reduced environmental impact (**?**). Autonomous vehicle technologies, though still evolving, promise to revolutionize transportation safety and accessibility in the coming decades.

2.4.1 AI in Education

AI has transformed educational settings by enhancing personalization, efficiency, and data-driven decision-making. Key applications include:

- Intelligent Tutoring Systems (ITS): These systems adapt content to individual student needs, improving learning outcomes. Research shows ITS can increase student performance by 0.3 to 1.0 standard deviations compared to traditional methods (**?**).
- Automated Assessment Tools: AI-driven grading systems provide instant feedback, reducing educator workload and enabling timely student support (**?**).
- Learning Analytics: AI analyzes student data to identify learning patterns, predict performance, and inform instructional strategies, with studies reporting improved retention rates (**?**).

These applications highlight AI's potential to address diverse learner needs, setting the stage for its use in specialized fields like financial management education.

2.4.2 Advantages of AI Across Sectors

AI offers several universal advantages that transcend specific industries. Enhanced efficiency stands out as a primary benefit, with organizations reporting productivity im-

provements of 11-40% following AI implementation (**?**). By automating routine tasks, AI frees human workers to focus on higher-value activities requiring creativity and emotional intelligence.

Improved decision-making represents another significant advantage. AI systems can process vast datasets beyond human capacity, identifying patterns and insights that inform strategic decisions. Organizations leveraging AI for decision support report 10-25% improvements in decision quality and consistency (**?**).

Personalization at scale has become possible through AI, allowing organizations to tailor experiences to individual preferences without prohibitive costs. This capability has transformed customer service, education, and content delivery across sectors (**?**).

Cost reduction through automation and optimization represents a compelling advantage for many organizations. Studies indicate that AI implementation can reduce operational costs by 15-25% in various business functions while improving service quality (**?**).

2.4.3 Disadvantages and Challenges of AI Implementation

Despite its benefits, AI implementation faces significant challenges across industries. Technical complexity and infrastructure requirements present substantial barriers, with organizations reporting that inadequate technical infrastructure hampers 40-60% of AI initiatives (**?**). Many organizations lack the computing resources, data architecture, and technical expertise necessary for successful AI deployment.

Ethical concerns and regulatory compliance pose increasingly important challenges. Issues of data privacy, algorithmic bias, and transparency affect AI applications across sectors (**?**). Studies indicate that 68% of consumers express concern about how their data is used in AI systems, highlighting the importance of ethical AI development (**?**).

Workforce disruption represents another significant challenge, with estimates suggesting that AI could automate 15-30% of existing jobs by 2030 (**?**). This disruption necessitates workforce retraining and adaptation, creating transitional challenges for organizations and educational institutions.

Implementation costs and uncertain ROI deter many organizations from AI adoption. Research indicates that 80-85% of AI projects fail to deliver expected returns, often due to implementation challenges, scope creep, or unrealistic expectations (**?**).

2.5 AI in Financial Management Education

2.5.1 Benefits of AI in Financial Management Education

Within the specific context of financial management education, AI offers transformative benefits for MBA students. Personalized learning is a significant advantage, as AI systems can adapt instructional content to match individual student needs, improving comprehension of complex financial theories (**?**). For instance, AI-driven platforms can adjust the pace of lessons on portfolio optimization based on a student's prior knowledge.

Efficiency in content delivery is another key benefit. AI tools, such as automated grading systems and virtual teaching assistants, reduce the administrative burden on instructors, allowing more time for interactive discussions on topics like mergers and acquisitions (**?**). This efficiency is critical in MBA programs where time constraints are common.

Enhanced decision-making skills also emerge from AI integration. Simulation-based AI tools enable students to practice financial forecasting and risk management in real-time scenarios, bridging the gap between theory and practice (**?**). Such experiential learning is invaluable for preparing students for the dynamic financial sector.

2.5.2 Challenges of AI in Financial Management Education

Despite its potential, AI integration presents notable challenges in educational settings. Technical limitations pose a significant barrier, as many educational institutions lack the infrastructure—such as high-speed internet or advanced computing resources—required to support AI systems effectively (**?**). This is particularly relevant in international settings like Beijing, where disparities in technological access may exist.

Ethical dilemmas also arise, including concerns over data privacy and algorithmic bias. Financial management education often involves sensitive data, and AI systems must ensure compliance with regulations like GDPR while avoiding biases that could skew financial analyses (**?**). These issues demand robust governance frameworks, which are often underdeveloped.

Resistance to change is a further challenge. Both students and faculty may exhibit skepticism toward AI tools, perceiving them as threats to traditional teaching methods or doubting their reliability in teaching nuanced topics like ethical investing (**?**). Overcoming this resistance requires targeted training and evidence of AI's efficacy.

2.5.3 Current Applications of AI in Financial Management Education

AI is increasingly applied in financial management education across global MBA programs. Adaptive learning platforms, such as those offered by Coursera, use AI to provide tailored courses on financial derivatives, adjusting content based on student performance (**?**). These platforms exemplify how AI can scale education to diverse learners.

AI-powered simulations are another prominent application. Tools like Bloomberg Terminal integrate AI to simulate market conditions, allowing students to test investment strategies without real-world risks (**?**). Such applications are particularly beneficial for international MBA students seeking practical exposure to global markets.

Chatbots and virtual assistants also play a role, offering real-time support for coursework queries on topics like capital budgeting (**?**). These tools enhance accessibility, especially for non-native English speakers at institutions like Beijing International Studies University, by providing multilingual assistance.

2.6 Previous Studies

Empirical research on AI in education has expanded rapidly, with several studies relevant to financial management. **?** conducted a mixed-methods study at a Latin American university, finding that AI-enhanced finance courses increased student engagement by 35% compared to traditional methods, attributed to interactive simulations and personalized feedback. However, the study focused on undergraduate rather than MBA contexts.

? reviewed AI applications across disciplines, noting a 40% improvement in learning outcomes when adaptive systems were used in business education. Their meta-analysis included finance-related courses but lacked specificity on MBA-level interventions, limiting its applicability to this study's focus.

In a study of technology acceptance, **?** applied TAM to pre-service teachers, finding that perceived usefulness strongly predicted adoption of e-learning tools ($\beta = 0.62$, $p < 0.01$). While insightful, this research did not address AI specifically or the unique needs of MBA students in finance.

? explored AI policy frameworks in higher education, identifying a lack of standardized guidelines for AI use in specialized fields like finance. Their qualitative analysis of 15 institutions revealed inconsistent adoption rates, suggesting a need for context-specific studies.

? investigated ChatGPT's impact on educational settings, reporting a 25% increase in

student comprehension of complex topics. However, the study's general focus excluded financial management specifics, leaving its relevance to MBA education unclear.

? examined LLMs in education, finding that 80% of surveyed educators valued their potential for content generation, such as case studies in finance. Yet, the study highlighted scalability issues in resource-constrained environments, a concern for international programs.

? analyzed AI-assisted learning in higher education, noting improved critical thinking skills among students using LLMs. Their experiment with 200 students showed a statistically significant improvement ($t(198) = 3.45$, $p < 0.001$), but financial management applications were not explored.

To clearly illustrate the research gap addressed by this study, Table **??** summarizes key prior studies, highlighting their limitations in addressing MBA-level financial management education for international students using LLMs and TAM.

Table 2.1 Summary of Previous Studies and Research Gap

Study	**MBA Focus**	**Financial Management**	**LLMs in Education**	**International Students**	**TAM Applied**
Córdova et al. **?**	×	✓	×	×	×
Gan et al. **?**	×	×	✓	×	×
Grassini **?**	×	×	✓	×	×
Laato et al. **?**	×	×	✓	×	×
Chan **?**	×	×	×	×	×
Chen et al. **?**	×	×	×	×	×
Teoh and Lim **?**	×	×	×	×	✓
This Study	✓	✓	✓	✓	✓

2.7 Summary

The literature reveals AI's substantial potential to enhance financial management education through personalization, efficiency, and practical skill development. Studies consistently demonstrate improved engagement and learning outcomes, with tools like simulations and adaptive platforms addressing key pedagogical needs. However, challenges such as technical limitations, ethical concerns, and resistance to change persist, often exacerbated in international contexts where resource disparities and cultural differences play a role.

A critical research gap emerges from this review: existing methods fail to adequately address the integration and acceptance of AI in financial management education for international MBA students, particularly those with diverse linguistic and academic backgrounds. Previous studies either focus on general education, undergraduate levels, or unrelated disciplines, neglecting the specific demands of graduate-level finance education in a globalized setting. Moreover, the lack of research applying TAM to AI adoption in this context leaves unanswered questions about the factors influencing its success among diverse learners.

Current approaches suffer from an inability to scale AI solutions across varied institutional capabilities and a failure to tailor interventions to the unique needs of international students, such as language support and culturally relevant content. This study addresses these shortcomings by investigating AI's effectiveness and acceptance in financial management education at Beijing International Studies University, aiming to develop a framework that enhances learning outcomes for a globally diverse MBA cohort.

CHAPTER 3 THEORETICAL FRAMEWORK

3.1 Research Motivation and Design

The integration of Artificial Intelligence (AI) tools, particularly Large Language Models (LLMs), into financial management education offers a promising solution to the challenges faced by international MBA students at Beijing International Studies University (BISU). These students encounter barriers such as language difficulties, diverse educational backgrounds, and varying technological familiarity, which traditional methods struggle to address effectively. While the Technology Acceptance Model (TAM) has been widely used to predict technology adoption, its application to AI tools in this specific multicultural context remains limited. This study extends TAM to include constructs relevant to AI and international education, providing a robust framework to evaluate acceptance and effectiveness.

Data will be collected from approximately 30 international MBA students at BISU enrolled in financial management courses. Participants, selected via purposive sampling, will have some exposure to AI tools (e.g., ChatGPT, DeepSeek) through coursework. An online survey, using a 7-point Likert scale, will measure TAM constructs. This approach ensures alignment with the study's focus on a diverse, relevant sample and supports the experimental and survey methods outlined in the subsequent chapters.

3.1.1 Research Model

Building on the theoretical foundations discussed in Chapter **??**, this study proposes an extended Technology Acceptance Model specifically tailored to examine the acceptance of AI tools in financial management education for international MBA students. The model incorporates constructs from TAM, TPB, and UTAUT, as well as factors specific to AI and cross-cultural educational contexts. Figure **??** presents the proposed research model.

The core of the model retains TAM's fundamental constructs: perceived usefulness, perceived ease of use, attitude toward using, behavioral intention, and actual usage. However, the model extends TAM by incorporating additional constructs that address the specific context of AI in international education:

- **Perceived Compatibility**: The degree to which AI tools are perceived as consistent

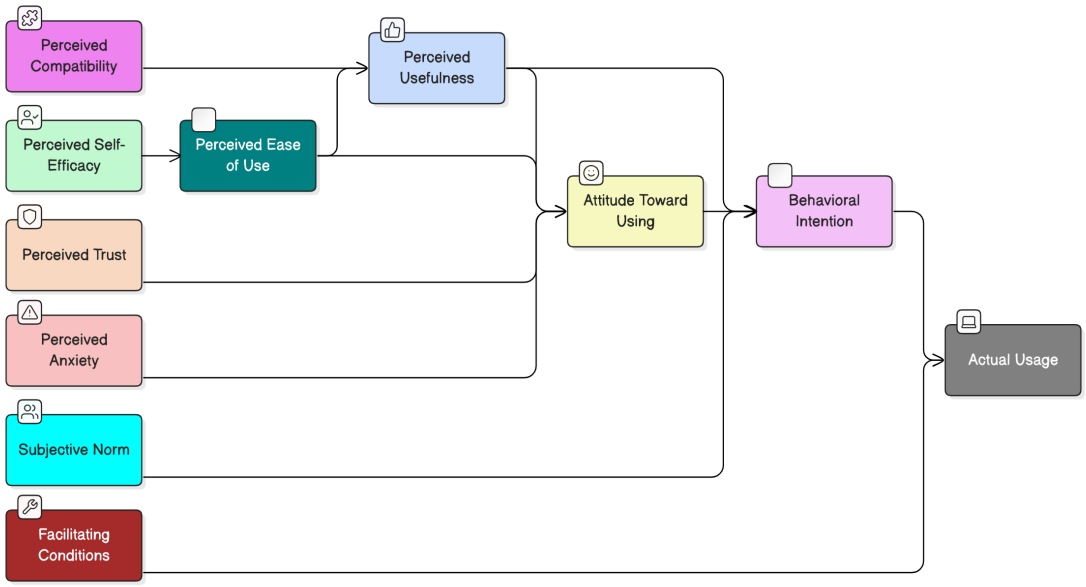

Figure 3.1 Extended Technology Acceptance Model fmor AI in Financial Management Education

with existing values, needs, and experiences of international MBA students (**?**).

- **Perceived Self-Efficacy**: Students' judgment of their capability to use AI tools effectively for financial management learning (**?**).
- **Perceived Trust**: The extent to which students believe that AI tools are reliable, accurate, and will protect their privacy (**?**).
- **Perceived Anxiety**: The degree of apprehension or fear when using or considering using AI tools (**?**).
- **Subjective Norm**: The perceived social pressure to use or not use AI tools, particularly from peers, instructors, and the institution (**?**).
- **Facilitating Conditions**: The degree to which students believe that organizational and technical infrastructure exists to support AI tool use (**?**).

This extended model acknowledges that acceptance of AI tools in financial management education is influenced not only by perceptions of usefulness and ease of use but also by compatibility with learning styles, self-efficacy in technology use, trust in AI systems, anxiety about AI, social influence, and institutional support. By incorporating these additional constructs, the model provides a more comprehensive framework for understanding the complex factors influencing AI acceptance among international MBA students.

3.1.2 Conceptual Foundations from Prior Research

The extended TAM draws on prior studies to justify its constructs. TAM's core constructs—perceived usefulness and ease of use—are well-established predictors of technology adoption in education (**???**). Research suggests students value AI tools that enhance performance and require minimal effort, particularly in complex fields like financial management (**?**).

Additional constructs are informed by recent research. Perceived compatibility highlights the importance of aligning AI tools with users' needs, as seen in studies on educational technology adoption (**?**). Self-efficacy is critical for students' confidence in using AI, with recent studies validating its role in AI tool acceptance (**?**). Trust is essential for AI adoption, as students need reliable and secure systems (**?**). Anxiety can deter engagement, particularly among those unfamiliar with AI, as evidenced in AI education research (**?**). Subjective norms and facilitating conditions, drawn from UTAUT, emphasize social and institutional influences on AI adoption (**?**). These foundations address the literature gap identified in Chapter **??**, tailoring TAM to AI in a multicultural MBA setting.

3.1.3 Hypotheses

Based on the extended Technology Acceptance Model presented above, this study proposes the following hypotheses regarding the acceptance of AI tools in financial management education for international MBA students at BISU.

Perceived Usefulness

Perceived usefulness has consistently been identified as a strong predictor of technology acceptance across various contexts (**?**). In educational settings, perceived usefulness refers to students' belief that using a technology will enhance their learning performance (**?**). For AI tools in financial management education, usefulness may manifest as improved understanding of complex concepts, enhanced problem-solving abilities, or more efficient learning processes.

Prior research supports the role of perceived usefulness in shaping attitudes and intentions toward technology use in education. Studies have found that students value technologies that improve academic outcomes, such as e-learning systems and mobile learning tools (**??**). For international MBA students, the perceived usefulness of AI tools may be particularly salient due to the complex nature of financial management and challenges like

language barriers and diverse backgrounds. AI tools offering personalized explanations or multilingual support may thus be seen as highly useful.

Based on these considerations, the following hypotheses are proposed:

H1a: Perceived usefulness positively influences attitude toward using AI tools in financial management education.

H1b: Perceived usefulness positively influences behavioral intention to use AI tools in financial management education.

Perceived Ease of Use

Perceived ease of use refers to the degree to which students believe that using AI tools will be free of effort (**?**). This construct is particularly relevant for technologies that may be perceived as complex or sophisticated, such as AI systems. Students who perceive AI tools as intuitive, accessible, and requiring minimal effort to learn and use are more likely to accept these tools.

Research has consistently demonstrated the influence of perceived ease of use on technology acceptance in educational contexts. Studies indicate that ease of use significantly shapes attitudes toward technology among educators and students, with user-friendly tools being more readily adopted (**??**). For international MBA students, ease of use may be especially important due to potential language barriers and varying levels of technological proficiency. AI tools with intuitive interfaces and multilingual capabilities may thus be more readily accepted.

Accordingly, the following hypotheses are proposed:

H2a: Perceived ease of use positively influences attitude toward using AI tools in financial management education.

H2b: Perceived ease of use positively influences perceived usefulness of AI tools in financial management education.

Attitude Toward Using

Attitude toward using refers to an individual's overall affective reaction to using a technology (**?**). In the context of AI tools for financial management education, attitude encompasses students' positive or negative feelings about incorporating these tools into their learning processes.

Prior studies have shown that attitude plays a mediating role between beliefs (such as perceived usefulness and ease of use) and behavioral intention. Research indicates

that attitude significantly influences students' intention to use educational technologies, with positive attitudes fostering greater adoption (**??**). For international MBA students, attitudes toward AI tools may be shaped by cultural factors, prior technology experiences, and perceptions of AI's role in finance.

Based on these considerations, the following hypothesis is proposed:

H3: Attitude toward using positively influences behavioral intention to use AI tools in financial management education.

Perceived Compatibility

Perceived compatibility, derived from Innovation Diffusion Theory (**?**), refers to the degree to which an innovation is perceived as consistent with existing values, needs, and experiences of potential adopters. In the context of AI tools for financial management education, compatibility encompasses alignment with students' learning styles, educational values, and prior technological experiences.

For international MBA students, compatibility may be particularly relevant due to diverse educational backgrounds and cultural values. AI tools that accommodate different learning approaches, respect cultural perspectives on education, and build upon familiar technological interfaces may be perceived as more compatible with students' needs and values.

Accordingly, the following hypothesis is proposed:

H4: Perceived compatibility positively influences perceived usefulness of AI tools in financial management education.

Perceived Self-Efficacy

Perceived self-efficacy, derived from Social Cognitive Theory (**?**), refers to individuals' judgment of their capability to use a technology effectively. In the context of AI tools for financial management education, self-efficacy encompasses students' confidence in their ability to interact with AI systems, interpret AI-generated content, and integrate AI tools into their learning processes.

For international MBA students, self-efficacy may vary based on prior exposure to advanced technologies, language proficiency, and confidence in navigating educational systems. Students with higher technological self-efficacy may perceive AI tools as easier to use and thus be more likely to accept these tools.

Based on these considerations, the following hypothesis is proposed:

H5: Perceived self-efficacy positively influences perceived ease of use of AI tools in financial management education.

Perceived Trust

Perceived trust refers to the extent to which individuals believe that a technology is reliable, accurate, and will protect their privacy and security (**?**). In the context of AI tools for financial management education, trust encompasses students' confidence in the accuracy of AI-generated content, the protection of their personal data, and the ethical use of their learning analytics.

For international MBA students, trust in AI tools may be influenced by cultural attitudes toward technology, awareness of AI capabilities and limitations, and institutional assurances regarding data protection. Students from cultures with higher uncertainty avoidance may place greater emphasis on trust when considering AI adoption.

Accordingly, the following hypothesis is proposed:

H6: Perceived trust positively influences attitude toward using AI tools in financial management education.

Perceived Anxiety

Perceived anxiety refers to the degree of apprehension, fear, or discomfort individuals feel when using or considering using a technology (**?**). In the context of AI tools for financial management education, anxiety may manifest as concerns about AI replacing human interaction, fear of making mistakes when using AI systems, or discomfort with AI's perceived complexity.

For international MBA students, anxiety related to AI tools may be exacerbated by language barriers, unfamiliarity with advanced technologies, or cultural attitudes toward AI. Students from cultures with lower technology adoption rates or those with limited prior exposure to AI may experience higher levels of anxiety.

Based on these considerations, the following hypothesis is proposed:

H7: Perceived anxiety negatively influences attitude toward using AI tools in financial management education.

3.1.4 Measurement Scales for Research Constructs

To operationalize the constructs in the extended Technology Acceptance Model, this study will develop measurement scales based on established instruments in the literature,

adapted to the specific context of AI tools in financial management education for international MBA students. The following sections detail the measurement items for each construct.

Perceived Usefulness

Perceived usefulness will be measured using a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree) with items adapted from **?** and validated in recent research on AI in education by **?**.

1. Using AI tools improves my performance in financial management courses.
2. Using AI tools enhances my effectiveness in learning financial concepts.
3. Using AI tools increases my productivity in completing financial management assignments.
4. Using AI tools makes it easier to understand complex financial theories.
5. Using AI tools helps me overcome language barriers in financial management education.
6. Overall, I find AI tools useful for my financial management studies.

Perceived Ease of Use

Perceived ease of use will be measured using a 7-point Likert scale with items adapted from **?** and validated in recent studies on AI educational tools by **?**.

1. Learning to use AI tools for financial management education is easy for me.
2. I find it easy to get AI tools to do what I want them to do.
3. My interaction with AI tools is clear and understandable.
4. I find AI tools flexible to interact with.
5. It is easy for me to become skillful at using AI tools.
6. Overall, I find AI tools easy to use for financial management learning.

Attitude Toward Using

Attitude toward using will be measured using a 7-point semantic differential scale with bipolar adjective pairs adapted from **?** and **?**, with recent validations by **?**:

1. Bad/Good
2. Foolish/Wise
3. Unfavorable/Favorable
4. Harmful/Beneficial

5. Negative/Positive
6. Unpleasant/Pleasant

The stem question will be: ”All things considered, using AI tools for financial management education is...”

Perceived Compatibility

Perceived compatibility will be measured using a 7-point Likert scale with items adapted from **?** and updated with insights from **?** on AI tool adoption:

1. Using AI tools is compatible with all aspects of my financial management studies.
2. Using AI tools fits well with my learning style.
3. Using AI tools is compatible with my cultural approach to education.
4. Using AI tools fits into my preferred way of learning financial concepts.
5. Using AI tools is consistent with my values regarding education.
6. Using AI tools complements the traditional teaching methods in my financial management courses.

Perceived Self-Efficacy

Perceived self-efficacy will be measured using a 7-point Likert scale with items adapted from **?** and validated in recent AI education research by **?**:

1. I could use AI tools for financial management learning if there was no one around to tell me what to do.
2. I could use AI tools for financial management learning if I had only the built-in help facility for assistance.
3. I could use AI tools for financial management learning if I had seen someone else using it before.
4. I could use AI tools for financial management learning if I had adequate time to complete the task.
5. I could use AI tools for financial management learning even if I had never used a similar system before.
6. I am confident in my ability to effectively use AI tools for financial management learning.

Perceived Trust

Perceived trust will be measured using a 7-point Likert scale with items adapted from **?** and supplemented with insights from **?** on trust in AI educational tools:

1. I believe that AI tools provide accurate information for financial management learning.
2. I believe that AI tools are reliable for financial management education.
3. I trust that AI tools will protect my personal data and privacy.
4. I believe that AI tools act in my best interest as a student.
5. I feel confident that AI tools will work as expected for financial management learning.
6. I trust the institution's implementation of AI tools for educational purposes.

Perceived Anxiety

Perceived anxiety will be measured using a 7-point Likert scale with items adapted from **?** and validated in recent AI research by **?**:

1. I feel apprehensive about using AI tools for financial management learning.
2. I fear that I might make mistakes that I cannot correct when using AI tools.
3. I hesitate to use AI tools for fear of making errors that I cannot fix.
4. AI tools for financial management learning are somewhat intimidating to me.
5. I worry that AI tools might provide incorrect information for my studies.
6. I feel uncomfortable about relying on AI for important financial management concepts.

Behavioral Intention and Actual Usage

Behavioral intention will be measured using a 7-point Likert scale with items adapted from **?** and **?**, with recent validations by **?**:

1. I intend to use AI tools for financial management learning in the next semester.
2. I predict that I will use AI tools for financial management learning regularly.
3. I plan to use AI tools for financial management learning frequently.
4. I intend to recommend AI tools to other students in financial management courses.

Actual usage will be measured through self-reported frequency and duration of use, as well as system logs where available, following methodologies in recent educational technology studies (**?**):

1. On average, how many times per week do you use AI tools for financial management learning? (1 = never, 7 = several times daily)
2. On average, how many hours per week do you spend using AI tools for financial management learning? (1 = none, 7 = more than 10 hours)
3. How would you characterize your usage of AI tools for financial management learning? (1 = non-use, 7 = heavy use)

Additionally, objective usage data will be collected from system logs, including frequency of access, duration of sessions, and types of interactions with the AI tools.

CHAPTER 4 RESEARCH METHODOLOGY

4.1 Problem Identification

The research questions guiding this study are:

1. To what extent do AI tools, specifically LLMs, enhance learning outcomes in financial management education for international MBA students?
2. What factors influence international MBA students' acceptance of AI tools in financial management education?
3. How do demographic factors (e.g., prior AI experience, language proficiency, cultural background) moderate the relationship between AI tool use and learning outcomes?

4.2 Research Framework

The research framework for this study integrates two theoretical perspectives: the Technology Acceptance Model (TAM) and the Input-Process-Output (IPO) model of learning. This integrated framework, illustrated in Figure **??**, provides a comprehensive approach to examining both the effectiveness of AI tools in enhancing learning outcomes and the factors influencing their acceptance among international MBA students.

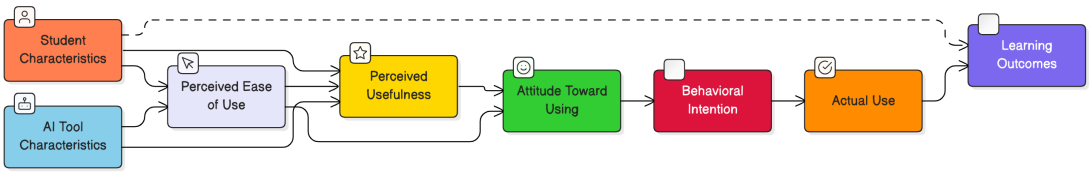

Figure 4.1 Research Framework

The framework incorporates the following components:

1. **Input Factors:** These include student characteristics (prior AI experience, language proficiency, cultural background) and AI tool characteristics (accessibility, responsiveness, accuracy). These factors are hypothesized to influence both the process variables and learning outcomes directly.
2. **Process Factors:** Derived from TAM, these include perceived ease of use, perceived usefulness, and attitude toward using AI tools. These factors mediate the relationship between input factors and behavioral intention to use AI tools.

3. **Output Factors:** These include behavioral intention, actual use of AI tools, and learning outcomes (quiz performance, engagement, satisfaction). The framework posits that actual use mediates the relationship between behavioral intention and learning outcomes.

4.3 Research Approaches and Methods

This study employs a mixed-methods approach, combining qualitative and quantitative research methods to provide a comprehensive understanding of AI's effectiveness and acceptance in financial management education. The mixed-methods design follows a sequential explanatory strategy (**?**), where quantitative data collection and analysis are followed by qualitative data collection and analysis to help explain and interpret the quantitative findings.

4.3.1 Qualitative Research

The qualitative component of this study employs semi-structured interviews and focus group discussions to gain in-depth insights into students' experiences with AI tools in financial management education. This approach is particularly valuable for understanding the nuanced ways in which AI tools influence learning processes and the contextual factors affecting their acceptance.

Semi-structured Interviews: Individual interviews will be conducted with a subset of participants (n = 15) selected through purposive sampling to represent diverse backgrounds and experiences. The interviews will explore students' perceptions of AI tools, challenges encountered during use, and perceived benefits for financial management learning. The interview protocol, informed by TAM constructs, will include questions such as:

- How would you describe your experience using AI tools for learning financial management concepts?
- What aspects of the AI tools did you find most useful for understanding complex financial concepts?
- What challenges did you encounter when using AI tools, and how did you overcome them?
- How did your cultural background or language proficiency influence your experience with AI tools?

Interviews will be audio-recorded, transcribed, and analyzed using thematic analysis to identify recurring patterns and themes related to AI effectiveness and acceptance.

Focus Group Discussions: Two focus group discussions (6-8 participants each) will be conducted to explore collective perspectives on AI tools in financial management education. These discussions will facilitate interaction among participants, allowing for the emergence of shared experiences and contrasting viewpoints. Focus groups will be particularly valuable for understanding social influences on AI acceptance and identifying common challenges and benefits across diverse student populations.

Qualitative data will be analyzed using NVivo software, employing both deductive coding based on TAM constructs and inductive coding to identify emergent themes. This analysis will provide rich contextual insights to complement and explain quantitative findings.

4.3.2 Quantitative Research

The quantitative component employs an experimental design and survey research to measure AI's impact on learning outcomes and identify factors influencing acceptance. This approach enables statistical testing of hypotheses derived from the research framework and generalization of findings to the broader population of international MBA students.

Experimental Design: A pre-test/post-test quasi-experimental design will be employed to assess the effectiveness of AI tools in enhancing financial management learning outcomes. The experimental procedure will follow these steps:

1. **Pre-test:** All participants will complete a financial management quiz (20 multiple-choice questions) covering key concepts in financial analysis, investment evaluation, and risk management. The quiz will be developed in consultation with financial management instructors and validated through pilot testing.
2. **Intervention:** Participants will be given 10 minutes to interact with their preferred AI tool (e.g., DeepSeek, ChatGPT, Grok, Gemini) to clarify concepts from the quiz and enhance their understanding. During this phase, participants' interactions with AI tools will be recorded (with consent) to analyze patterns of use.
3. **Post-test:** Participants will complete the same financial management quiz to measure improvements in performance. The time taken to complete the quiz will also be recorded as a measure of efficiency.

The experimental design includes several control variables, such as prior financial

knowledge, language proficiency, and demographic factors, to account for potential confounding effects. The primary outcome measures will include:

- **Performance Improvement:** Calculated as the difference between post-test and pre-test scores, indicating the extent to which AI tools enhanced understanding of financial concepts.
- **Time Efficiency:** Measured as the difference in time taken to complete the post-test compared to the pre-test, indicating whether AI tools improved the speed of problem-solving.
- **Engagement Metrics:** Derived from recordings of AI interactions, including the number of questions asked, types of questions (clarification, elaboration, application), and patterns of interaction.

Survey Research: A comprehensive survey will be administered to all participants to measure factors influencing AI acceptance based on the extended TAM framework. The survey will include validated scales for TAM constructs, adapted from previous research (**??**), as well as items measuring student characteristics and perceptions of AI tool characteristics. Key constructs measured will include:

- **Perceived Usefulness (PU):** Measured using a 7-point Likert scale with items such as ”Using AI tools enhances my understanding of financial management concepts” and ”AI tools improve my efficiency in solving financial problems.”
- **Perceived Ease of Use (PEOU):** Assessed with items such as ”Learning to use AI tools for financial management is easy for me” and ”I find it easy to get AI tools to do what I want them to do.”
- **Attitude Toward Using (ATU):** Measured with items such as ”Using AI tools for learning financial management is a good idea” and ”I enjoy using AI tools for financial management education.”
- **Behavioral Intention (BI):** Assessed with items such as ”I intend to use AI tools for learning financial management in the future” and ”I would recommend AI tools to other students learning financial management.”
- **Student Characteristics:** Including prior AI experience (frequency and types of use), language proficiency (self-rated and standardized test scores), and cultural background (individualism-collectivism scale, power distance preferences).
- **AI Tool Characteristics:** Measured through items assessing perceived accessibility, responsiveness, and accuracy of AI tools used during the intervention.

The survey will be administered online using Qualtrics, with measures taken to ensure data quality, including attention checks and reverse-coded items. Survey data will be analyzed using structural equation modeling (SEM) to test the relationships proposed in the research framework (**?**). SEM is particularly appropriate for this study as it allows for simultaneous testing of multiple relationships and the inclusion of latent variables representing TAM constructs.

Data Analysis Techniques: Quantitative data analysis will employ several statistical techniques:

- **Descriptive Statistics:** Means, standard deviations, and frequencies will be calculated for all variables to provide an overview of the sample characteristics and response patterns.
- **Paired t-tests:** To compare pre-test and post-test scores, assessing the statistical significance of performance improvements after AI tool use.
- **Correlation Analysis:** To examine bivariate relationships between variables, providing preliminary insights into the associations between student characteristics, TAM constructs, and learning outcomes.
- **Structural Equation Modeling (SEM):** To test the full research model, including direct and indirect effects, mediation, and moderation relationships. SEM will be conducted using AMOS software, with model fit assessed using standard indices (CFI, TLI, RMSEA, SRMR).
- **Moderation Analysis:** To examine how demographic factors (e.g., language proficiency, cultural background) moderate the relationship between AI tool use and learning outcomes, using multi-group SEM and interaction terms.

4.4 Sampling Design

The study employs a purposive sampling approach to ensure representation of diverse international MBA students at Beijing International Studies University. The sampling design considers several factors to enhance the validity and generalizability of findings.

Target Population: The target population comprises international MBA students enrolled in financial management courses at BISU. This population is characterized by diversity in cultural backgrounds, language proficiency, prior educational experiences, and exposure to AI technologies.

Sample Size: To detect a medium-to-large effect size (Cohen's $d = 0.6$) with 80%

power and $\alpha = 0.05$ in paired t-tests comparing pre-test and post-test scores in financial management quizzes, a minimum sample size of approximately 28 participants is required. For structural equation modeling (SEM) to evaluate the extended Technology Acceptance Model, the proposed model is simplified to approximately 3 parameters, requiring a minimum of 30 observations. To mitigate potential attrition and non-response, robust retention strategies, such as participant engagement protocols and follow-up reminders, will be implemented.

Sampling Procedure: Participants will be recruited through several channels:

- Announcements in financial management course group and other online groups
- Snowball sampling, where initial participants refer peers from different cultural backgrounds

To ensure representation of diverse perspectives, stratified sampling will be employed based on key demographic characteristics:

- **Geographic Region:** Ensuring representation from major regions (East Asia, South Asia, Africa, Europe, Americas)
- **Language Proficiency:** Including students with varying levels of English proficiency
- **Prior AI Experience:** Incorporating both experienced and novice AI users

Inclusion Criteria: Participants must meet the following criteria:

- Currently enrolled in the MBA program at BISU
- International student status
- Currently taking or have completed at least one financial management course
- Basic familiarity with digital technologies (able to use a computer and access online resources)

Exclusion Criteria: Students will be excluded if they:

- Have not taken any financial management courses
- Are unable to participate in both the pre-test and post-test sessions
- Have extensive professional experience in financial management (to avoid ceiling effects)

Sampling for Qualitative Components: For the qualitative phase, a subset of participants will be selected using maximum variation sampling to ensure diverse perspectives. Selection criteria will include:

- Performance change (both high and low improvers)

- Cultural background (representing different regions)
- AI experience (both experienced and novice users)
- Language proficiency (varying levels)

Ethical Considerations: The sampling process will adhere to ethical guidelines for research with human participants. Informed consent will be obtained from all participants, with clear explanations of the study purpose, procedures, and data usage. Participation will be voluntary, with the option to withdraw at any point without consequences. To protect privacy, all data will be anonymized, with participants assigned unique identifiers.

4.5 Implementation Plan and Timeline

The research will be executed in a systematic manner following a detailed implementation plan with specific milestones. The entire research process will be divided into four distinct phases:

4.5.1 Phase 1: Preparation and Instrument Development

In this initial phase, the focus is on laying the groundwork for the study. A comprehensive literature review will be conducted to refine the understanding of recent advancements in AI in education, with an emphasis on analyzing the learning challenges faced by international students in financial management. This review will synthesize key findings on technology acceptance in educational contexts and document the theoretical frameworks and methodological approaches that will guide the research.

Following the literature review, the development of research instruments will commence. A financial management pre-test and post-test quiz will be created in collaboration with subject matter experts to ensure accuracy and relevance. This quiz will be pilot-tested with two students to assess its validity and reliability, and it will be refined based on feedback and item analysis. Additionally, Technology Acceptance Model (TAM) survey instruments will be adapted from validated scales, and interview and focus group protocols will be developed to gather qualitative data. Data collection forms will also be created to record interactions with AI tools.

Ethical considerations will be addressed by obtaining necessary permissions from MBA program administrators. Informed consent forms and information sheets will be developed to ensure participants understand the study's purpose and their rights, and data management protocols will be established to safeguard privacy and confidentiality. Re-

cruitment preparation will involve developing materials such as announcements, posters, and group messages, along with creating a participant database and tracking record sheets to manage recruitment and scheduling.

Deliverables for Phase 1: Validated research instruments, recruitment materials, and database records sheets.

4.5.2 Phase 2: Participant Recruitment and Quantitative Data Collection

This phase focuses on recruiting participants and collecting quantitative data. A recruitment campaign will be launched through multiple channels to attract international MBA students who meet the study's criteria. Potential participants will be screened based on inclusion and exclusion criteria, and informed consent will be obtained from those who qualify. Eligible participants will then be scheduled for experimental sessions.

During the experimental sessions, participants will complete a pre-test in a controlled environment to establish their baseline knowledge of financial management concepts. They will then engage with AI tools, with their interactions recorded for analysis. A post-test will be administered under standardized conditions to measure improvements in understanding, and any observations or technical issues will be documented. Concurrently, an online TAM survey will be distributed to all participants to evaluate their perceptions of the AI tools. Reminders will be sent to maximize response rates, and data quality and completeness will be monitored throughout the survey period. After the survey closes, preliminary data processing will begin, including scoring the pre-test and post-test quizzes, cleaning and coding survey responses, organizing AI interaction recordings, and preparing datasets for statistical analysis.

Deliverables for Phase 2: Complete quantitative dataset (pre-test/post-test scores, survey responses, interaction metrics) from 30+ participants.

4.5.3 Phase 3: Qualitative Data Collection and Analysis

The qualitative phase seeks to gain deeper insights into participants' experiences with AI tools. Based on preliminary quantitative results, a diverse subset of participants will be selected using maximum variation sampling and invited to participate in semi-structured interviews and focus groups. These sessions will be scheduled to collect qualitative data efficiently.

Fifteen individual interviews, each lasting 45 to 60 minutes, will be conducted to

explore participants' personal experiences in detail. These interviews will be audio-recorded, transcribed, and supplemented with notes on non-verbal cues and contextual information. Preliminary coding of early interviews will begin to identify emerging patterns. Additionally, two focus group discussions, each with 6 to 8 participants, will be facilitated to capture collective perspectives and document group dynamics and interaction patterns. These discussions will also be recorded and transcribed.

Qualitative data analysis will involve importing transcripts into NVivo software and developing an initial coding framework based on TAM constructs. First-cycle coding, using descriptive and in vivo techniques, will be conducted, followed by second-cycle coding with pattern and axial methods to refine themes. Key themes and illustrative quotes will be identified, and member checking with selected participants will be performed to validate the findings.

Deliverables for Phase 3: Coded qualitative dataset, thematic analysis report, and member checking validation.

4.5.4 Phase 4: Comprehensive Data Analysis and Reporting

In the final phase, the emphasis is on analyzing data and sharing results. Quantitative data analysis will include conducting descriptive statistical analysis, performing paired t-tests to compare pre-test and post-test scores, and calculating correlation matrices for key variables. Structural equation models will be developed and tested to evaluate the TAM framework, with moderation analyses conducted to assess the influence of demographic factors. Bootstrapping techniques will be used to validate these models.

An integrated mixed-methods analysis will triangulate quantitative and qualitative findings, identifying areas of convergence and divergence. Qualitative data will be used to explain quantitative patterns, leading to the formulation of evidence-based recommendations for AI integration in financial management education. A comprehensive research report will be prepared, accompanied by an executive summary and presentation materials. Stakeholder engagement will involve presenting findings to BISU faculty, sharing results with participants, and developing an implementation plan based on the study's outcomes.

Deliverables for Phase 4: Final research report, presentations, and implementation recommendations.

Figure**??** below provides a comprehensive overview of the entire implementation process:

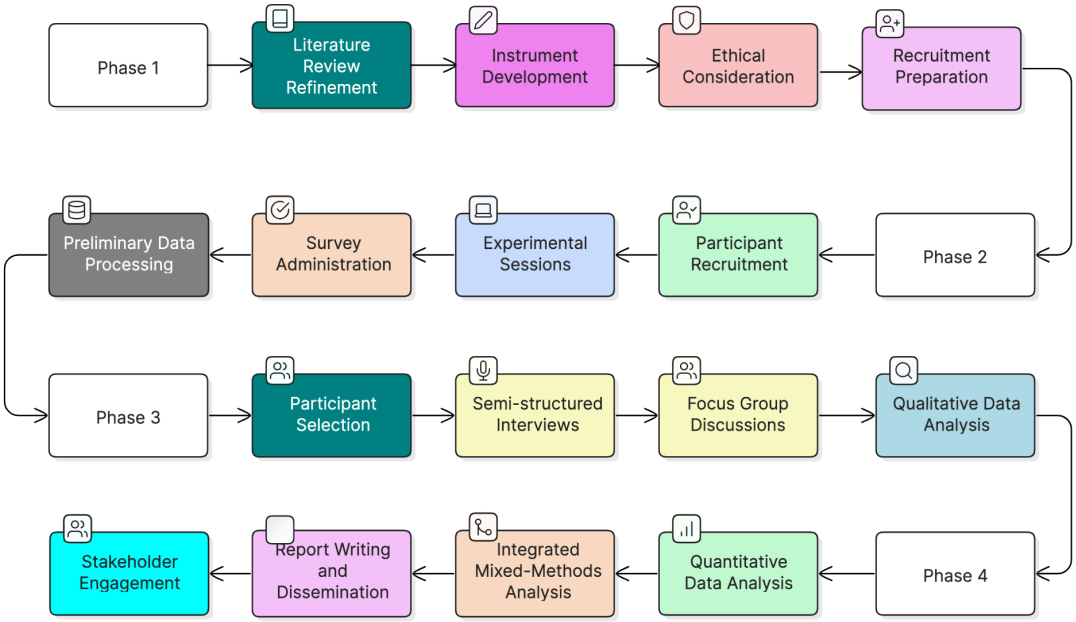

Figure 4.2 Overview of the Implementation Process

4.5.5 Quality Assurance and Risk Management

Throughout all phases, the following quality assurance and risk management strategies will be implemented:

- **Data Quality Checks:** Regular audits of collected data to identify and address missing or inconsistent information.
- **Methodological Triangulation:** Using multiple methods to enhance validity and reliability of findings.
- **Peer Debriefing:** Regular consultations with colleagues not involved in the study to challenge assumptions and interpretations.
- **Contingency Planning:** Development of alternative strategies for potential challenges (e.g., recruitment difficulties, technical issues with AI tools).
- **Ethical Monitoring:** Continuous attention to ethical considerations, particularly regarding data privacy

CHAPTER 5 RESULTS AND ANALYSIS

5.1 Overview

This chapter reports the empirical findings from the mixed-methods study on AI-supported financial management education for international MBA students at BISU. It presents quantitative results from the pre-test/post-test experiment, qualitative insights from interviews and interaction logs, and the outcomes of the extended TAM analyses.

5.2 Data Preparation and Sample Description

This section describes the data cleaning steps, missing data handling, and the final sample characteristics (e.g., demographics, prior AI experience, English proficiency, baseline financial knowledge). It also summarizes instruments and variables used in subsequent analyses.

5.3 Quantitative Results

5.3.1 Pre-test vs. Post-test Performance

Reports descriptive statistics, effect sizes, and inferential tests (e.g., paired t-test or Wilcoxon signed-rank) comparing pre- and post-test scores. Includes confidence intervals and visualizations as appropriate.

5.3.2 Time Efficiency

Compares completion times between pre- and post-test phases and assesses statistical significance of differences. Discusses implications for learning efficiency.

5.3.3 Differential Impact Across Subgroups

Examines performance and time improvements across subgroups (e.g., language proficiency, baseline knowledge, prior AI experience) using appropriate analyses (e.g., ANCOVA, interaction terms).

5.4 Qualitative Results

5.4.1 Interaction Patterns with AI

Presents thematic analysis of AI interaction logs and interview/focus group data, highlighting strategies, question types, and engagement patterns associated with better outcomes.

5.5 TAM Measurement Model

5.5.1 Reliability and Convergent/Divergent Validity

Reports Cronbach's alpha, composite reliability, average variance extracted (AVE), and discriminant validity checks (e.g., HTMT) for all constructs.

5.5.2 Measurement Model Fit

Summarizes confirmatory factor analysis results and model fit indices (e.g., CFI, TLI, RMSEA, SRMR), with brief commentary on adequacy of the measurement model.

5.6 TAM Structural Model

5.6.1 Path Coefficients and Hypothesis Testing

Presents standardized path coefficients, significance levels, R^2 values, and tests of the study's hypotheses.

5.6.2 Moderation and Robustness Checks

Reports moderation analyses (e.g., cultural dimensions, prior experience) and robustness checks (e.g., alternative specifications, multicollinearity diagnostics).

5.7 Summary of Findings

Synthesizes the main results across quantitative and qualitative analyses, noting where findings converge or diverge. This summary sets up the discussion and conclusions in Chapter **??**.

CHAPTER 6 CONCLUSION

This thesis proposal has outlined a comprehensive study to investigate the effectiveness of AI tools, particularly Large Language Models, in enhancing the learning outcomes of international MBA students in financial management education at Beijing International Studies University. The research addresses a significant gap in the literature regarding the application of AI in specialized business education for diverse student populations.

The study employs a mixed-methods approach, combining experimental research to assess learning outcomes and survey research based on the Technology Acceptance Model (TAM) to understand adoption factors. This dual focus provides a holistic understanding of both the effectiveness of AI tools and the factors influencing their acceptance among international MBA students.

The theoretical framework integrates TAM with cultural dimensions and student characteristics, offering a comprehensive model for understanding technology acceptance in multicultural educational settings. The methodology includes a pre-test/post-test experimental design and a survey measuring TAM constructs, providing robust data for analysis.

Anticipated findings suggest that AI tools will significantly enhance learning outcomes in financial management education, with variations based on student characteristics and interaction patterns. The TAM analysis is expected to reveal key factors influencing AI adoption, including cultural dimensions and prior experience.

6.1 Limitations

Despite its comprehensive design, this research has several limitations that should be acknowledged:

- **Sample Limitations:** The study focuses on international MBA students at BISU, which may limit the generalizability of findings to other educational contexts or student populations. The sample size, while sufficient for statistical analysis, may not capture the full diversity of international students.
- **Short-term Intervention:** The experimental design involves a relatively short-term interaction with AI tools (10 minutes), which may not fully capture the potential long-term benefits of sustained AI use in educational settings.
- **Self-reported Measures:** The TAM survey relies on self-reported measures, which

may be subject to social desirability bias or limited self-awareness regarding technology acceptance factors.

6.2 Future Research Directions

This study opens several promising avenues for future research:

- **Longitudinal Studies:** Future research could employ longitudinal designs to examine the long-term impact of AI tools on learning outcomes and skill development in financial management education.
- **Comparative Analysis:** Comparative studies across different cultural contexts, educational institutions, or AI tools could provide broader insights into the effectiveness and acceptance of AI in business education.
- **Ethical Considerations:** In-depth exploration of ethical issues related to AI use in education, including privacy concerns, algorithmic bias, and academic integrity, represents an important direction for future research.
- **AI Literacy Development:** Studies on how to develop AI literacy among MBA students and faculty could enhance the effective use of these tools in educational settings.

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